Chapter 1 — Differentiation: Last-Minute Revision

Maharashtra HSC · Mathematics & Statistics Part II · File 2 of 3 · Read this the night before

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1 The chapter on one page

Differentiation measures **rate of change**: $\frac{dy}{dx}$ is how fast y changes per unit change in x, and geometrically it is the **slope of the tangent**. This chapter is about *techniques* for awkward functions. There are only ten question types (T1–T10) plus three proofs. Learn to recognise the type; the method then follows automatically.

Exam Tip: Board facts (from 6 papers, Mar 2024 – Jun 2026)

- Q1(v) MCQ (2 marks) came from this chapter **every time**.
- Section D (4 marks) had a **proof + "hence"** in 5 of 6 papers: **parametric proof ×4, chain-rule proof ×1** (plus chain rule proof once as a 3-mark Q21).
- Other repeats: implicit $\sqrt{x} + \sqrt{y} = \sqrt{a}$ style (2 marks), log differentiation (2 marks), n^{th}

derivative / second-order "show that" (3 marks), inverse-trig simplification (3 marks).
- Pattern from past papers only — not a guarantee.

2 Formula sheet

Formula: Basic derivatives

y	y'	y	y'
x^n	nx^{n-1}	$\sec x$	$\sec x \tan x$
$\sin x$	$\cos x$	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\cos x$	$-\sin x$	$\cot x$	$-\operatorname{cosec}^2 x$
$\tan x$	$\sec^2 x$	e^x	e^x
$\log x$	$\frac{1}{x}$	a^x	$a^x \log a$
$\log_a x$	$\frac{1}{x \log a}$	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$

Formula: Inverse trig derivatives

y	y'	y	y'
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$	$\cot^{-1} x$	$-\frac{1}{1+x^2}$
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$	$\operatorname{cosec}^{-1} x$	$-\frac{1}{x\sqrt{x^2-1}}$

"Co" partner = negative. Conditions: $|x| < 1$ for $\sin^{-1}, \cos^{-1}$; $|x| > 1$ for $\sec^{-1}, \operatorname{cosec}^{-1}$.

Formula: The six rules of this chapter

1. **Chain:** $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$
2. **Inverse:** $\frac{dx}{dy} = \frac{1}{dy/dx}$, $(f^{-1})'(k) = \frac{1}{f'(a)}$ where $f(a) = k$
3. **Log diff.:** $\frac{d}{dx} f^g = f^g \left[\frac{g}{f} f' + g' \log f \right]$
4. **Implicit:** differentiate every term; each y-term picks up $\frac{dy}{dx}$
5. **Parametric:** $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$; $\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \cdot \frac{dt}{dx}$
6. u **w.r.t.** v: $\frac{du}{dv} = \frac{du/dx}{dv/dx}$

Formula: n^{th} derivatives

y	y_n
$(ax+b)^m$	$m(m-1)\cdots(m-n+1)a^n(ax+b)^{m-n}$
$\dfrac{1}{ax+b}$	$\dfrac{(-1)^n n!\,a^n}{(ax+b)^{n+1}}$
$\log(ax+b)$	$\dfrac{(-1)^{n-1}(n-1)!\,a^n}{(ax+b)^n}$
e^{ax+b}	$a^n e^{ax+b}$
a^{px+q}	$p^n(\log a)^n a^{px+q}$
$\sin(ax+b)$ / $\cos(ax+b)$	$a^n \sin\left(ax+b+\frac{n\pi}{2}\right)$ / $a^n \cos\left(ax+b+\frac{n\pi}{2}\right)$
$e^{ax}\sin(bx+c)$ / $e^{ax}\cos(bx+c)$	$r^n e^{ax}\sin(bx+c+n\alpha)$ / $r^n e^{ax}\cos(bx+c+n\alpha)$; $r=\sqrt{a^2+b^2}$, $\alpha=\tan^{-1}\frac{b}{a}$

Formula: Trig substitutions (for T4)

See	Put	Becomes
$\dfrac{2x}{1-x^2}, \dfrac{2x}{1+x^2}, \dfrac{1-x^2}{1+x^2}$	$x=\tan\theta$	$\tan 2\theta, \sin 2\theta, \cos 2\theta$
$\sqrt{1-x^2}, 2x\sqrt{1-x^2}, 3x-4x^3$	$x=\sin\theta$	$\cos\theta, \sin 2\theta, \sin 3\theta$
$2x^2-1, 4x^3-3x$	$x=\cos\theta$	$\cos 2\theta, \cos 3\theta$
$\sqrt{\dfrac{a-x}{a+x}}$	$x=a\cos 2\theta$	$\tan\theta$
$\dfrac{(p+q)x}{1-pqx^2}$	split	$\tan^{-1}px+\tan^{-1}qx$
$\dfrac{a\cos x+b\sin x}{\sqrt{a^2+b^2}}$	$\dfrac{a}{\sqrt{a^2+b^2}}=\cos\alpha$	$\cos(x-\alpha)$
$\dfrac{\cos x}{1\pm\sin x}, \sqrt{\dfrac{1\pm\cos x}{1\mp\cos x}}$	half-angles	$\tan\left(\frac{\pi}{4}\mp\frac{x}{2}\right)$, $\cot\frac{x}{2}$ / $\tan\frac{x}{2}$

Always write the range of θ you assume.

Question looks like…	Type	Do this

3 Which method? — recognise in 5 seconds

Question looks like…	Type	Do this
function inside function, or values table of f, g, f', g'	**T1 Chain**	outside$'$ $\times$ inside$'$
"derivative of inverse", $(f^{-1})'(k)$	**T2 Inverse**	find a with $f(a) = k$, answer $\frac{1}{f'(a)}$
$\sin^{-1}$, $\tan^{-1}$ … of simple inside	**T3 Inv-trig direct**	formula $\times$ inner derivative
$\sin^{-1}$, $\tan^{-1}$ … of messy algebra / trig	**T4 Substitution**	substitute, collapse, then differentiate
variable in base **and** exponent; long product	**T5 Log diff.**	take log, differentiate, multiply by y
x and y mixed in one equation	**T6 Implicit**	differentiate, collect y'
$x = f(t), y = g(t)$	**T7 Parametric**	$\frac{dy/dt}{dx/dt}$
"differentiate u w.r.t. v"	**T8**	$\frac{du/dx}{dv/dx}$
$\frac{d^2y}{dx^2}$, "show that … $y_2 + \ldots y_1 \ldots = 0$"	**T9 2nd order**	find y_1, clear fractions, differentiate again
"n^{th} derivative"	**T10**	$y_1, y_2, y_3 \to$ pattern / table

4 Worked examples by type (Easy → Medium → Hard)

4.1 T1 — Chain rule

Worked Example: E: $y = (x^3 - 2x - 1)^5$

$$y' = 5(x^3 - 2x - 1)^4 \cdot (3x^2 - 2)$$

Worked Example: M: $y = e^{3\sin^2 x - 2\cos^2 x}$

$$y' = e^{3\sin^2 x - 2\cos^2 x} \cdot (6\sin x \cos x + 4\cos x \sin x) = e^{3\sin^2 x - 2\cos^2 x} \cdot 5\sin 2x$$

($\frac{d}{dx}(-2\cos^2 x) = -4\cos x(-\sin x) = +4\sin x \cos x$ — watch that sign.)

Worked Example: H: value table (Ex. 1.1 Q4)

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	1	6	-3	4
4	3	4	5	-6
6	5	2	-4	7

(i) $r(x) = f[g(x)]$: $r'(2) = f'[g(2)] \cdot g'(2) = f'(6) \cdot 4 = (-4)(4) = -16$.

(ii) $R(x) = g[3 + f(x)]$: $R'(4) = g'[3 + f(4)] \cdot f'(4) = g'(6) \cdot 5 = 7 \cdot 5 = 35$.

Method: write the chain-rule formula **first**, then plug in from the table — inside values come from the $f(x)/g(x)$ columns, derivatives from the f'/g' columns.

Worked Example: H (trick): $y = 25^{\log_5(\sec x)} - 16^{\log_4(\tan x)}$

$25^{\log_5 \sec x} = (5^{\log_5 \sec x})^2 = \sec^2 x$ and $16^{\log_4 \tan x} = \tan^2 x$. So $y = \sec^2 x - \tan^2 x = 1$ and $y' = 0$. **Simplify before differentiating.**

4.2 T2 — Derivative of inverse function

Worked Example: E: $y = \sqrt{x}$ via inverse

$x = y^2 \Rightarrow \frac{dx}{dy} = 2y \Rightarrow \frac{dy}{dx} = \frac{1}{2y} = \frac{1}{2\sqrt{x}}$.

Worked Example: M: inverse of $y = x^5 + 2x^3 + 3x$ at $x = 1$

$\frac{dy}{dx} = 5x^4 + 6x^2 + 3$; at $x = 1$ it is 14. So $\frac{dx}{dy} = \frac{1}{14}$.

Worked Example: H: $f(x) = x^3 + 3x + 1$, find $(f^{-1})'(5)$

Need a with $f(a) = 5$: $a^3 + 3a - 4 = 0 \Rightarrow a = 1$ $(1 + 3 - 4 = 0)$. $f'(x) = 3x^2 + 3$, $f'(1) = 6$. $(f^{-1})'(5) = \frac{1}{6}$. **Trap:** $f'(5)$ is wrong — 5 is a y-value.

4.3 T3 — Inverse trig, direct

Worked Example: E: $y = \cot^{-1}(4^x)$

$$y' = -\frac{1}{1 + (4^x)^2} \cdot 4^x \log 4 = -\frac{4^x \log 4}{1 + 16^x}$$

Worked Example: M: $y = \cos^3[\cos^{-1}(x^3)]$

$\cos(\cos^{-1} x^3) = x^3$ for $|x| \leq 1$, so $y = (x^3)^3 = x^9$ and $y' = 9x^8$.

Worked Example: H: $y = \operatorname{cosec}^{-1}(e^{-x})$, $x < 0$

$$y' = -\frac{1}{e^{-x}\sqrt{e^{-2x} - 1}} \cdot (-e^{-x}) = \frac{1}{\sqrt{e^{-2x} - 1}}$$

4.4 T4 — Inverse trig by substitution

Worked Example: E: $y = \tan^{-1}\sqrt{\dfrac{1+\cos x}{1-\cos x}}, 0 < x < \pi$

$\dfrac{1+\cos x}{1-\cos x} = \dfrac{2\cos^2\frac{x}{2}}{2\sin^2\frac{x}{2}} = \cot^2\frac{x}{2}$, so the root is $\cot\frac{x}{2} = \tan\left(\frac{\pi}{2} - \frac{x}{2}\right)$. $y = \frac{\pi}{2} - \frac{x}{2} \Rightarrow y' = -\frac{1}{2}$.

Worked Example: M: $y = \cos^{-1}(3x - 4x^3), |x| < \frac{1}{2}$

Put $x = \sin\theta, \theta \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$: $3x - 4x^3 = \sin 3\theta = \cos\left(\frac{\pi}{2} - 3\theta\right)$ with $\frac{\pi}{2} - 3\theta \in (0, \pi)$. $y = \frac{\pi}{2} - 3\theta = \frac{\pi}{2} - 3\sin^{-1}x \Rightarrow y' = -\dfrac{3}{\sqrt{1-x^2}}$.

Worked Example: H: $y = \tan^{-1}\sqrt{\dfrac{3-x}{3+x}}$ **[Board Jul 2024, 3 marks]**

Put $x = 3\cos 2\theta, \theta \in (0, \frac{\pi}{2})$: $\sqrt{\frac{3-x}{3+x}} = \sqrt{\frac{2\sin^2\theta}{2\cos^2\theta}} = \tan\theta$. $y = \theta = \frac{1}{2}\cos^{-1}\frac{x}{3} \Rightarrow y' = \frac{1}{2}\cdot\frac{-1}{\sqrt{1-\frac{x^2}{9}}}\cdot\frac{1}{3} = -\dfrac{1}{2\sqrt{9-x^2}}$.

4.5 T5 — Logarithmic differentiation

Worked Example: E: $y = x^{\tan^{-1}x}$

$\log y = \tan^{-1}x \cdot \log x \Rightarrow \dfrac{y'}{y} = \dfrac{\tan^{-1}x}{x} + \dfrac{\log x}{1+x^2}$.

$$y' = x^{\tan^{-1}x}\left[\frac{\tan^{-1}x}{x} + \frac{\log x}{1+x^2}\right]$$

Worked Example: M: $y = (\sin x)^x$

$\log y = x\log\sin x \Rightarrow \dfrac{y'}{y} = x\cot x + \log\sin x \Rightarrow y' = (\sin x)^x\,[x\cot x + \log\sin x]$.

Worked Example: H: $y = [(\tan x)^{\tan x}]^{\tan x}$ **at** $x = \frac{\pi}{4}$ **(Ex. 1.3 Q2(viii))**

$y = (\tan x)^{\tan^2 x} \Rightarrow \log y = \tan^2 x \log\tan x$.

$$\frac{y'}{y} = 2\tan x\sec^2 x\log\tan x + \tan^2 x\cdot\frac{\sec^2 x}{\tan x} = \sec^2 x\tan x(2\log\tan x + 1)$$

At $x = \frac{\pi}{4}$: $\tan = 1, \sec^2 = 2, \log 1 = 0, y = 1$. So $y' = 1\cdot 2\cdot 1\cdot(0+1) = 2$.

4.6 T6 — Implicit functions

Worked Example: E: $\sqrt{x}+\sqrt{y}=\sqrt{a}$ [Board Feb 2026]

$\dfrac{1}{2\sqrt{x}}+\dfrac{y'}{2\sqrt{y}}=0\Rightarrow y'=-\sqrt{\dfrac{y}{x}}$. (Twin: $x\sqrt{x}+y\sqrt{y}=a\sqrt{a}\Rightarrow y'=-\sqrt{\frac{x}{y}}$ [Board Jul 2025].)

Worked Example: M: $\cos(xy)=x+y$

$-\sin(xy)(xy'+y)=1+y'\Rightarrow -y'[1+x\sin(xy)]=1+y\sin(xy)$

$$y'=-\frac{1+y\sin(xy)}{1+x\sin(xy)}$$

Worked Example: H: $\log_{10}\left(\dfrac{x^3-y^3}{x^3+y^3}\right)=2$, show $y'=-\dfrac{99x^2}{101y^2}$

Remove the log first: $\dfrac{x^3-y^3}{x^3+y^3}=10^2=100\Rightarrow x^3-y^3=100x^3+100y^3\Rightarrow 101y^3=-99x^3$.

Differentiate: $303y^2y'=-297x^2\Rightarrow y'=-\dfrac{297x^2}{303y^2}=-\dfrac{99x^2}{101y^2}$.

Memory Trick: Shortcuts you may use to check (show steps in the exam)

- $x^my^n=(x+y)^{m+n}\Rightarrow y'=\frac{y}{x}$.
- $F\left(\frac{x^n\pm y^n}{x^n\mp y^n}\right)=\text{const}\Rightarrow y'=\frac{y}{x}$ (componendo–dividendo).
- Infinite chain: replace the repeating tail by y — e.g. $y=\sqrt{\cos x+y}\Rightarrow y^2=\cos x+y\Rightarrow y'=\frac{-\sin x}{2y-1}=\frac{\sin x}{1-2y}$.

4.7 T7 — Parametric

Worked Example: E: $x=a\cot\theta, y=b\operatorname{cosec}\theta$

$$\frac{dy}{dx}=\frac{-b\operatorname{cosec}\theta\cot\theta}{-a\operatorname{cosec}^2\theta}=\frac{b}{a}\cdot\frac{\cot\theta}{\operatorname{cosec}\theta}=\frac{b}{a}\cos\theta$$

Worked Example: M: $x=\operatorname{cosec}^2\theta, y=\cot^3\theta$ at $\theta=\frac{\pi}{6}$

$\frac{dx}{d\theta}=-2\operatorname{cosec}^2\theta\cot\theta$, $\frac{dy}{d\theta}=-3\cot^2\theta\operatorname{cosec}^2\theta$. $\frac{dy}{dx}=\frac{3}{2}\cot\theta$; at $\frac{\pi}{6}$: $\frac{3}{2}\cdot\sqrt{3}=\dfrac{3\sqrt{3}}{2}$.

Worked Example: H: $x=e^{\sin 3t}, y=e^{\cos 3t}$, show $\dfrac{dy}{dx}=-\dfrac{y\log x}{x\log y}$

$\frac{dx}{dt}=3\cos 3t\,e^{\sin 3t}=3x\cos 3t$, $\frac{dy}{dt}=-3\sin 3t\,e^{\cos 3t}=-3y\sin 3t$. $\frac{dy}{dx}=-\frac{y\sin 3t}{x\cos 3t}$. Since $\log x=\sin 3t$ and $\log y=\cos 3t$: $\dfrac{dy}{dx}=-\dfrac{y\log x}{x\log y}$.

4.8 T8 — One function w.r.t. another

Worked Example: E: 3^x w.r.t. $\log_x 3$

$v = \log_x 3 = \frac{\log 3}{\log x} \Rightarrow \frac{dv}{dx} = -\frac{\log 3}{x(\log x)^2}$; $\frac{du}{dx} = 3^x \log 3$.

$$\frac{du}{dv} = \frac{3^x \log 3}{-\frac{\log 3}{x(\log x)^2}} = -3^x\, x(\log x)^2$$

Worked Example: M: x^x w.r.t. $x^{\sin x}$

$$\frac{du}{dv} = \frac{x^x(1+\log x)}{x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right)}$$

Worked Example: H: $\tan^{-1}\frac{x}{\sqrt{1-x^2}}$ w.r.t. $\sec^{-1}\frac{1}{2x^2-1}$, $0<x<1$

u: $x = \sin\theta \Rightarrow \frac{x}{\sqrt{1-x^2}} = \tan\theta \Rightarrow u = \theta = \sin^{-1} x$. $v = \cos^{-1}(2x^2-1)$: $x = \cos\phi$, $\phi \in (0, \frac{\pi}{2}) \Rightarrow v = \cos^{-1}(\cos 2\phi) = 2\phi = 2\cos^{-1} x$. $\frac{du}{dv} = \frac{1/\sqrt{1-x^2}}{-2/\sqrt{1-x^2}} = -\frac{1}{2}$.

4.9 T9 — Second order

Worked Example: E: $y = e^{2x}\sin 3x$

$y_1 = e^{2x}(2\sin 3x + 3\cos 3x)$ $y_2 = e^{2x}[2(2\sin 3x + 3\cos 3x) + (6\cos 3x - 9\sin 3x)] = e^{2x}(12\cos 3x - 5\sin 3x)$.

Worked Example: M: $x = a\cos\theta, y = b\sin\theta$, find $\frac{d^2y}{dx^2}$

$\frac{dy}{dx} = \frac{b\cos\theta}{-a\sin\theta} = -\frac{b}{a}\cot\theta$.

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(-\frac{b}{a}\cot\theta\right)\cdot\frac{d\theta}{dx} = \frac{b}{a}\operatorname{cosec}^2\theta\cdot\frac{1}{-a\sin\theta} = -\frac{b}{a^2}\operatorname{cosec}^3\theta$$

Worked Example: H: $y = \log\left(x+\sqrt{x^2+a^2}\right)$, show $(x^2+a^2)y_2 + xy_1 = 0$

$y_1 = \frac{1}{x+\sqrt{x^2+a^2}}\left(1+\frac{x}{\sqrt{x^2+a^2}}\right) = \frac{1}{\sqrt{x^2+a^2}} \Rightarrow \sqrt{x^2+a^2}\,y_1 = 1$. Differentiate: $\sqrt{x^2+a^2}\,y_2 + \frac{x}{\sqrt{x^2+a^2}}y_1 = 0$. Multiply by $\sqrt{x^2+a^2}$: $(x^2+a^2)y_2 + xy_1 = 0$.

4.10 T10 — n^{th} derivative

Worked Example: E: $y = e^{ax+b}$

$y_1 = ae^{ax+b}$, $y_2 = a^2e^{ax+b}$, …, $y_n = a^ne^{ax+b}$.

Worked Example: M: $y = \dfrac{1}{x}$

$y_1 = -x^{-2}, y_2 = (-1)(-2)x^{-3} = \frac{2!}{x^3}, y_3 = -\frac{3!}{x^4} \Rightarrow y_n = \dfrac{(-1)^n n!}{x^{n+1}}.$

Worked Example: H: $y = \log(2x+3)$ **(and** $\log x$ **— Board Feb 2025)**

$y_1 = \frac{2}{2x+3}, y_2 = \frac{-2^2}{(2x+3)^2}, y_3 = \frac{(-1)^2\,2!\,2^3}{(2x+3)^3} \Rightarrow$

$$y_n = \frac{(-1)^{n-1}(n-1)!\,2^n}{(2x+3)^n}$$

With $a = 1, b = 0$: $(\log x)_n = \dfrac{(-1)^{n-1}(n-1)!}{x^n}.$

5 The three proofs — skeletons to memorise

Proof: P1 Chain rule: $y = f(u), u = g(x) \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

1. Increments $\delta x, \delta u, \delta y$ (all $\neq 0$); $\dfrac{\delta y}{\delta x} = \dfrac{\delta y}{\delta u} \times \dfrac{\delta u}{\delta x}.$
2. Take $\lim_{\delta x \to 0}$; $\delta u \to 0$ **as** $\delta x \to 0$ **since** u **is continuous** … (I)
3. $\lim_{\delta u \to 0} \frac{\delta y}{\delta u} = \frac{dy}{du}$, $\lim_{\delta x \to 0} \frac{\delta u}{\delta x} = \frac{du}{dx}$ … (II)
4. R.H.S. exists & finite ⇒ L.H.S. exists ⇒ y differentiable and $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ … (III)

Proof: P2 Parametric: $x = f(t), y = g(t), \frac{dx}{dt} \neq 0 \Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

1. Increments $\delta t, \delta x, \delta y$ (all $\neq 0$); $\dfrac{\delta y}{\delta x} = \dfrac{\delta y/\delta t}{\delta x/\delta t}.$
2. Take $\lim_{\delta t \to 0}$; $\delta x \to 0$ **as** $\delta t \to 0$ **since** x **is continuous** … (I)
3. $\lim \frac{\delta y}{\delta t} = \frac{dy}{dt}$, $\lim \frac{\delta x}{\delta t} = \frac{dx}{dt} \neq 0$ … (II)
4. R.H.S. exists & finite ⇒ $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ … (III)

Proof: P3 Inverse: $\frac{dx}{dy} = \frac{1}{dy/dx}$

1. $\dfrac{\delta x}{\delta y} \cdot \dfrac{\delta y}{\delta x} = 1 \Rightarrow \dfrac{\delta x}{\delta y} = \dfrac{1}{\delta y/\delta x}.$
2. Limit $\delta x \to 0$; $\delta y \to 0$ since y is continuous.
3. $\lim \frac{\delta y}{\delta x} = \frac{dy}{dx} \neq 0 \Rightarrow \frac{dx}{dy} = \frac{1}{dy/dx}.$

Typical "hence" parts asked: $x = at^2, y = 2at \Rightarrow \frac{1}{t}$; $y = at^2, x = 2at \Rightarrow t$; $x = \sin t, y = \cos t \Rightarrow -\tan t$; 7^x w.r.t. $x^7 \Rightarrow \frac{7^x \log 7}{7x^6}$; $\frac{d}{dx}\frac{1}{\sqrt{\sin x}} = -\frac{\cos x}{2(\sin x)^{3/2}}.$

6 Frequently confused pairs

Confusion	Truth
$\sin^{-1} x$ vs $(\sin x)^{-1}$	$\sin^{-1} x$ is the inverse function; $(\sin x)^{-1} = \operatorname{cosec} x$
$\sin^2 x$ vs $\sin x^2$	$(\sin x)^2 \to 2\sin x \cos x$; $\sin(x^2) \to 2x \cos x^2$
$\frac{d}{dx}(x^n)$ vs $\frac{d}{dx}(a^x)$ vs $\frac{d}{dx}(x^x)$	power rule / exponential rule / **log differentiation**
$\frac{dy}{dx}$ parametric	$\frac{dy/dt}{dx/dt}$, **not** $\frac{dx/dt}{dy/dt}$
$\frac{d^2y}{dx^2}$ parametric	$\frac{d}{dt}(y') \times \frac{dt}{dx}$, **not** $\frac{y''(t)}{x''(t)}$
$(f^{-1})'(k)$	$\frac{1}{f'(a)}$ where $f(a) = k$ — **not** $\frac{1}{f'(k)}$
$(\log x)_n$ vs $\left(\frac{1}{x}\right)_n$	$\frac{(-1)^{n-1}(n-1)!}{x^n}$ vs $\frac{(-1)^n n!}{x^{n+1}}$
$\log_x a$	rewrite as $\frac{\log a}{\log x}$ before differentiating

7 Exam precautions

Common Mistake: Top mistakes that lose marks

1. Missing inner derivative in chain rule.
2. Missing $\frac{dy}{dx}$ on y-terms in implicit differentiation; forgetting product rule on xy, x^2y.
3. Taking log of a **sum** — split $y = u + v$ first.
4. Forgetting to multiply by y (and substitute it back) in log differentiation.
5. In substitution questions, not converting θ back to x (e.g. $\theta = \tan^{-1} x$) before differentiating.
6. Skipping the continuity reason in proofs — that line carries marks.
7. "At a point" questions: not computing the missing coordinate first (e.g. $y = 1$ at $x = 0$ for $\log(x + y) = xy$).
8. MCQs: write **option letter + answer** (some papers give zero for the letter alone).

8 Rapid-revision checklist

Tick each one only if you can do it **without looking**.

- ☐ Write all 12 basic + 6 inverse-trig derivatives in 2 minutes
- ☐ Prove the chain rule (P1) with all three numbered equations
- ☐ Prove the parametric rule (P2) and do a "hence" part
- ☐ Prove the inverse rule (P3)
- ☐ Solve a values-table chain-rule MCQ (T1-H)
- ☐ Find $(f^{-1})'(k)$ correctly (T2-H)
- ☐ Recognise and apply the 7 trig substitutions (T4)
- ☐ Log-differentiate f^g and a sum of two such terms (T5)
- ☐ Implicit: $\sqrt{x} + \sqrt{y} = \sqrt{a}$, $x\sqrt{x} + y\sqrt{y} = a\sqrt{a}$, $\log(x + y) = xy$ at $x = 0$ (T6)
- ☐ Parametric first **and** second derivative (T7, T9-M)
- ☐ Differentiate u w.r.t. v (T8)

☐ "Show that" second order for $\sin^{-1} x$, $\cos(m\cos^{-1} x)$, $A\cos(\log x) + B\sin(\log x)$ (T9)

☐ n^{th} derivatives of $\log x$, $\frac{1}{x}$, $\sin(ax+b)$, $e^{ax}\cos(bx+c)$ (T10)

Note: Gaps and limitations

This file groups Ex. 1.1–1.5 and Misc. Ex. 1 into ten types; every textbook exercise question falls into one of them. Rare items not drilled here: Misc. Ex. (II)(1) piecewise-function composites (same chain rule, but check which piece applies at the point) and Misc. 5(vi) $\frac{d^2x}{dy^2} = -\left(\frac{dy}{dx}\right)^{-3}\frac{d^2y}{dx^2}$ — see File 1.